

A Literal Fixed-Shift Generic-Affine Packet

Exact TT^* , Signed-Prefix Crosswalks, and a Growability Audit

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Abstract

TPC-108 isolated the generic-affine arithmetic object, while TPC-123–125 specify the archive, physical reassembly, and prescribed-shift interfaces needed to call it without replacing the literal coefficient. We give a self-contained invocation theorem for one complete fixed- h_0 packet. For every archived block

$$T_b = \sum_{z \in I_b} A_b(z) e(-\alpha_b z), \quad s_b u_b - a_b d_b = h_0,$$

we prove the exact lag identity

$$|T_b|^2 = \sum_r e(-\alpha_b r) \sum_{z, z+r \in I_b} A_b(z+r) \overline{A_b(z)}$$

and the exact canonical-order Abel identity. For the complete block family $S_X = \sum_b \gamma_b T_b$, the normalized literal ratios obey

$$\frac{|S_X|}{\sum_b |\gamma_b| M_b} \leq \left(\sum_b q_b \frac{|T_b|^2}{M_b^2} \right)^{1/2}, \quad q_b = \frac{|\gamma_b| M_b}{\sum_c |\gamma_c| M_c}.$$

This identifies a precise growing theorem that would close the H3 subgate after its separately audited outer loss. The present paper proves the finite identities and their typed invocation only. It does not prove the required uniform bound, a fixed-shift L2 saving, a parity breakthrough, or a twin-prime result.

1 Why another block statement is needed

The hard packet of TPC-15 is an exact scalar remainder whose $o(X)$ estimate is equivalent, within that decomposition, to the corresponding weighted fixed-shift prime-pair asymptotic [8]. TPC-93 exports its low window into affine coordinates while preserving content, projector, polarization, and the extracted Möbius factors [9]. TPC-107 insists that the signed filter be applied before positive majorization [2]; TPC-108 then writes the exact TT^* kernel and explains why an unrestricted replacement would reach a Chowla-strength problem [1, 3].

Those identities are not callable merely because a sequence with the same support has been constructed. TPC-119 found that the audited snapshot lacked a complete native reconnection archive [4]. TPC-123–125 therefore separate three interfaces: path conservation, leaf-to-physical cover, and direct prescribed- h_0 intertwining [5–7]. This paper starts *after* those interfaces. Its purpose is to state exactly what the extracted arithmetic block is, what finite equalities may be checked, and what genuinely growing assertion is still missing.

Claim boundary. An exact finite equality is L0. Its verified attachment to the literal fixed- h_0 packet is L1. Only a uniform coefficient-specific estimate on the complete growing family can be L2. A finite computation, a frozen form, an averaged shift, or a surrogate sign model does not change this typing.

2 The typed literal block

Fix one prescribed nonzero integer h_0 . For each scale X , let $\mathcal{B}_X$ be the *complete* set of long generic blocks selected by the verified archive and physical-cover maps. A block record is

$$b = (I_b, d_b, s_b, u_b, a_b, \alpha_b, \gamma_b, \mathbf{m}_b, \mathbf{p}_b, \mathbf{n}_b),$$

where:

(i) $I_b \subset \mathbb{Z}$ is a finite ordered interval;

(ii) $D_b(z) = d_b + s_b z$, $V_b(z) = u_b + a_b z$, and

$$s_b u_b - a_b d_b = h_0;$$

(iii) $\alpha_b \in \mathbb{R}/\mathbb{Z}$ is the literal frequency, and $\gamma_b \in \mathbb{C}$ is the literal outer multiplier;

(iv) $\mathbf{m}_b, \mathbf{p}_b, \mathbf{n}_b$ contain, respectively, all masks, the path/provenance key, and the unique inherited physical normalization.

The coefficient is not abbreviated to its two visible Möbius factors. We write

$$A_b(z) = \mu(D_b(z))\mu(V_b(z))\omega_b(z)\mathbf{1}_{\mathcal{M}_b}(z), \quad (1)$$

where ω_b is the full complex reconstruction weight and $\mathbf{1}_{\mathcal{M}_b}$ is the product of the actual content, coprimality, prefix, interval, projector, and polarization masks. In particular, the extracted factor $\mu(B)\mathbf{1}_{(B, V_b(z))=1}$ from TPC-93 is part of (1); it may not be deleted.

Definition 2.1 (Archive extraction). Let c_X be the native coefficient vector, M_X the path-expanded archive matrix of TPC-123, and E_b the row selector and exact reconstruction map for block b . The block is *literally extracted* if

$$(A_b(z))_{z \in I_b} = E_b M_X c_X \quad (2)$$

with equality of values and of the metadata listed above.

Remark 2.2. Equation (2) is stronger than agreement of one scalar block sum. Null pairs can preserve a scalar sum while changing the lag kernel, a prefix, or a determinant fiber. It is also stronger than equality after absolute values, which erases the signed channel isolated in TPC-107.

Define

$$e(t) = e^{2\pi i t}, \quad T_b = \sum_{z \in I_b} A_b(z) e(-\alpha_b z), \quad M_b = \sum_{z \in I_b} |A_b(z)|.$$

Zero-mass blocks are retained in the archive but omitted from ratios.

3 Exact TT^* and prefix identities

Theorem 3.1 (Literal lag expansion). *For every finite literal block,*

$$|T_b|^2 = \sum_{r \in I_b - I_b} e(-\alpha_b r) C_b(r), \quad C_b(r) = \sum_{\substack{z \in I_b \\ z+r \in I_b}} A_b(z+r) \overline{A_b(z)}. \quad (3)$$

Moreover $C_b(-r) = \overline{C_b(r)}$, so the right side is real and nonnegative even though its individual lag terms need not be.

Proof. Expanding the square gives

$$|T_b|^2 = \sum_{x,z \in I_b} A_b(x) \overline{A_b(z)} e(-\alpha_b(x-z)).$$

Set $r = x - z$, hence $x = z + r$, and collect all admissible pairs. The symmetry follows by conjugating and replacing r by $-r$. Nonnegativity follows from the left side, not from termwise positivity. $\square$

Remark 3.2 (The four-Möbius interior). Substituting (1) into $C_b(r)$ exposes

$$\mu(D_b(z+r))\mu(V_b(z+r))\mu(D_b(z))\mu(V_b(z))$$

times the literal masks and weights. Both shifted affine pairs still have determinant h_0 . Replacing this expression by independent random signs, or dropping the mask intersection at $z, z+r$, proves a statement about another kernel.

The ordered form supplies a second exact diagnostic. Write $I_b = \{z_1 < \dots < z_m\}$, and suppose one declared factorization is

$$A_b(z_i) e(-\alpha_b z_i) = \sigma_i w_i.$$

The choice of σ_i and w_i , and the order z_i , are part of the archive. Put $P_k = \sum_{i \leq k} \sigma_i$.

Theorem 3.3 (Canonical-order Abel crosswalk). *For every such factorization,*

$$T_b = P_m w_m + \sum_{k=1}^{m-1} P_k (w_k - w_{k+1}). \quad (4)$$

Consequently

$$|T_b| \leq \max_{1 \leq k \leq m} |P_k| \left(|w_m| + \sum_{k < m} |w_k - w_{k+1}| \right).$$

Proof. Since $\sigma_i = P_i - P_{i-1}$, with $P_0 = 0$,

$$\sum_{i=1}^m (P_i - P_{i-1}) w_i = P_m w_m + \sum_{i=1}^{m-1} P_i (w_i - w_{i+1}).$$

The inequality is the triangle inequality. $\square$

Remark 3.4. Theorems 3.1 and 3.3 are two exact views of the same literal block, not two interchangeable hypotheses. TT^* asks for a weighted aggregate of four-point lags. Abel asks for every signed prefix under one canonical order. A bound in one language must be transferred with all literal weights before it can be used in the other.

4 Complete-family normalization

The physical generic contribution has the exact form

$$S_X = \sum_{b \in \mathcal{B}_X} \gamma_b T_b. \quad (5)$$

Set

$$\mathcal{A}_X = \sum_b |\gamma_b| M_b, \quad q_b = \frac{|\gamma_b| M_b}{\mathcal{A}_X}$$

when $\mathcal{A}_X > 0$. Then (q_b) is a probability vector.

Theorem 4.1 (Literal family invocation). *For the complete archived block family,*

$$\frac{|S_X|}{\mathcal{A}_X} \leq \sum_{b:M_b>0} q_b \frac{|T_b|}{M_b} \leq \left(\sum_{b:M_b>0} q_b \frac{|T_b|^2}{M_b^2} \right)^{1/2}. \quad (6)$$

Both inequalities have sharp finite examples.

Proof. The first inequality follows from

$$|S_X| \leq \sum_b |\gamma_b| |T_b| = \mathcal{A}_X \sum_b q_b |T_b| / M_b.$$

The second is Cauchy–Schwarz in the probability measure q_b . Equality in both occurs when all nonzero $\gamma_b T_b$ have a common phase and all ratios $|T_b|/M_b$ are equal. $\square$

Corollary 4.2 (Typed termwise sufficient condition). *Assume, on the complete actual growing family and in one physical normalization, that*

$$\sum_{b:M_b>0} q_b \frac{|T_b|^2}{M_b^2} \leq X^{-2\eta+o(1)}. \quad (7)$$

If the separately proved outer reconstruction and norm-conversion cost is $X^{\lambda_{\text{out}}+o(1)}$, then the net relative saving delivered by (6) is

$$X^{-(\eta-\lambda_{\text{out}})+o(1)}.$$

In the deliberately termwise TPC-108 ledger, this certificate reaches the stronger displayed linear generic-window exponent when

$$\eta - \lambda_{\text{out}} \geq \frac{1}{200}. \quad (8)$$

This is a conservative sufficient condition for that linear window, not the exact quadratic-to-amplitude threshold of MVP1. Calling the MVP1 packet saving 1/400 still requires the separately verified physical TT^ crosswalk and prefactor.*

Proof. Take the square root in (7), apply (6), and multiply by the separately certified outer cost. The last condition is the stronger termwise linear-window threshold used in TPC-108; it is not derived from a finite block and is not claimed necessary for every coherent H3 proof. $\square$

No hidden averaging. The quantifier in (7) is over the actual X -indexed family. It cannot be replaced by any of the following:

- (a) a fixed finite list of blocks;
- (b) an average over shifts when h_0 is prescribed;
- (c) an average over unrelated phases or moduli;
- (d) a density-one subfamily without charging its complement;
- (e) a family selected after inspecting favorable phase cancellation.

The archive rule, the block family, and any good/bad selection must be predeclared.

5 What finite audits can and cannot decide

Proposition 5.1 (Finite saturation). *For every finite support I , every nonnegative magnitudes $(m_z)_{z \in I}$, and every phase α , there are coefficients with $|A(z)| = m_z$ such that*

$$\left| \sum_{z \in I} A(z) e(-\alpha z) \right| = \sum_{z \in I} m_z.$$

There are also coefficients with the same magnitudes whose sum is zero whenever the polygon inequality $\max m_z \leq \frac{1}{2} \sum m_z$ holds.

Proof. For saturation take $A(z) = m_z e(\alpha z + \theta)$. For cancellation, the polygon inequality is the classical necessary and sufficient condition that vectors of lengths m_z close to a polygon; rotate those vectors back by $e(\alpha z)$. $\square$

Thus support, occupancy, and magnitudes alone do not determine the signed ratio. Conversely, one unusually small finite ratio does not prove (7). A finite experiment is valuable for checking the literal extraction and choosing a candidate analytic lemma, but not for certifying an asymptotic exponent.

Definition 5.2 (Four verdicts for this interface). The block audit uses:

GO-INTERFACE	The complete finite archive-to-block extraction, metadata, normalization, TT^* , and prefix identities pass. This is at most L1.
GO-H3	In addition, a theorem proves (7) on the complete actual growing family and the separate cost satisfies (8). Only this can close the declared H3 subgate.
STOP-DECLARED-ROUTE	An exact growing counterfamily inside the quantifiers of a declared candidate lemma violates it. This stops that lemma, not all signed routes.
NOT-TESTABLE	The archive extraction, complete block census, metadata, physical normalization, or growing theorem is absent.

6 Executable certificate and status

The standard-library script `experiments/tpc126_literal_packet_audit.py` checks on exact integer metadata and ordinary double-precision finite data:

1. the determinant identity and the presence of a declared sample metadata subset, not completeness of the literal archive metadata;
2. the complex lag identity and its Hermitian symmetry;
3. the canonical Abel identity and BV bound;
4. both family inequalities and a saturation example;
5. the logical separation between GO-INTERFACE and GO-H3.

It writes a deterministic JSON certificate. It does not sample primes, estimate a growing exponent, or promote a finite regression to arithmetic evidence.

Statement	Level and present status
Lag expansion and Abel identity	Exact finite theorem, L0.
Archive-to-literal block equality	Callable L1 interface; requires the actual TPC-123–125 artifacts.
Complete growing bound (7)	Open coefficient-specific arithmetic input.
Outer loss λ_{out}	Separate physical theorem/ledger input; unknown costs are not zero.
H3 fixed- h_0 gate	NOT-TESTABLE from the current artifacts; no L2 theorem.

Conclusion. The literal arithmetic target is now an executable object with an exact normalization and a sharp family-level invocation rule. This is forward movement at the interface: a future theorem has one explicit left-hand side, one complete set of quantifiers, and one loss threshold. The hard step remains the growing signed estimate itself. Nothing here proves Möbius cancellation for the actual family, breaches the parity barrier, gives a prime-pair lower bound, establishes a Hardy–Littlewood asymptotic, or proves the twin-prime conjecture.

References

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